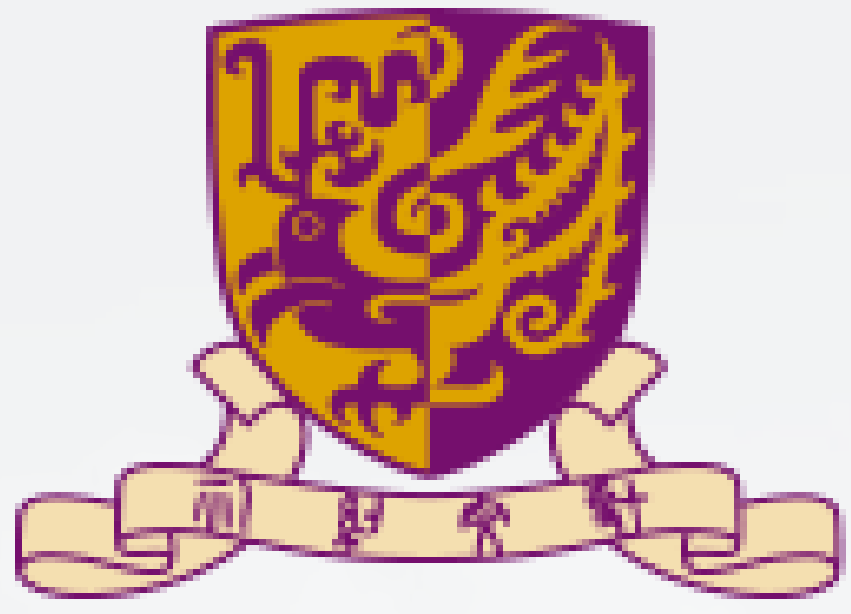




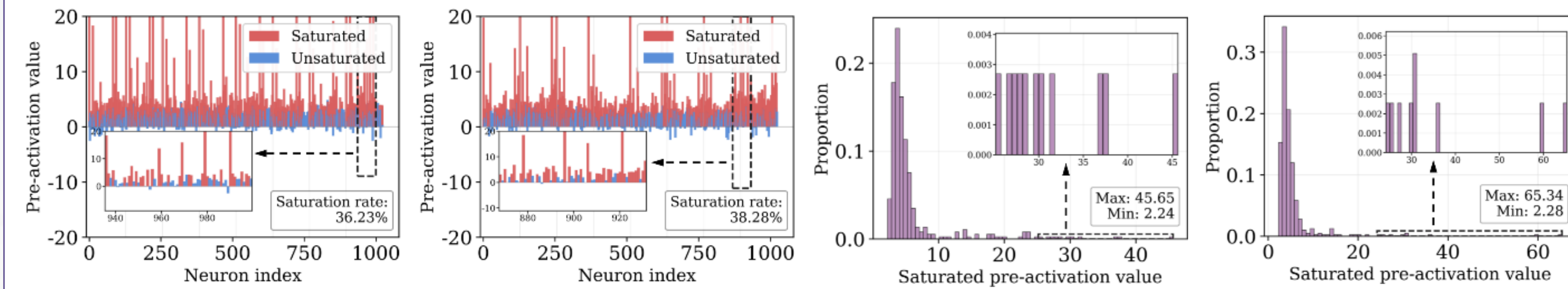
SmoothSpike: Spiking Transformer with Learnable Hadamard Transformation

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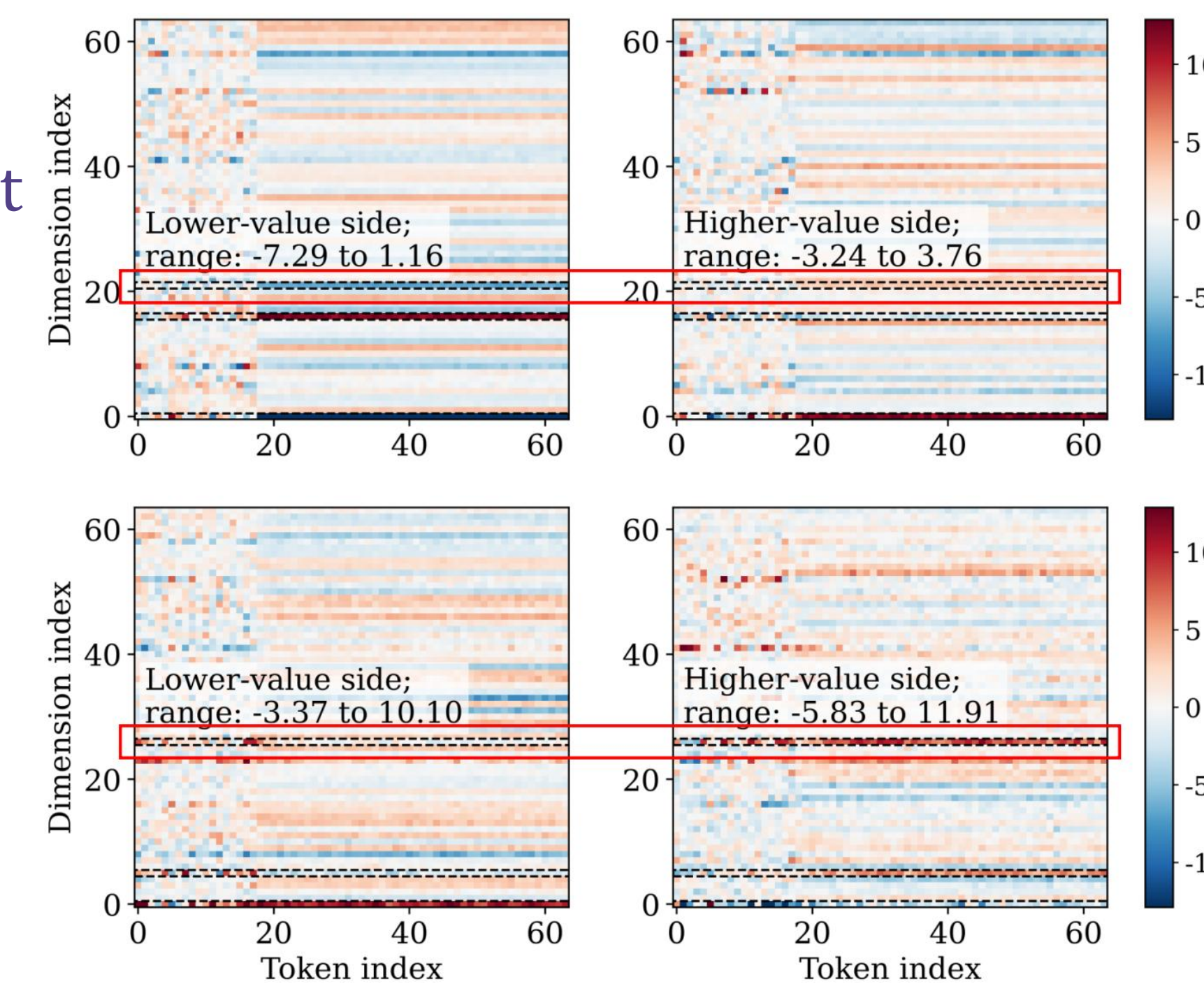
Motivation



- Spike Saturation Problem.** For distinct input currents $x_1 > x_2 \gg 0$, a spiking neuron may fire at the maximum rate for both inputs:
 $\sum_{t=1}^T SN(x_1, V_{t-1}^{(1)}) = \sum_{t=1}^T SN(x_2, V_{t-1}^{(2)}) = T$. Thus, different inputs produce identical spike trains.
- Using Spikingformer-12-768 on CoLA, we observe that many neurons across layers reach the maximum firing rate. Although their pre-activation inputs span a wide dynamic range, saturated neurons map them to the same spike outputs.
- This causes **spike saturation-induced information homogenization**, where diverse semantic signals become indistinguishable. Reducing spike saturation is therefore crucial for preserving representation capacity in SNNs.

Learnable Orthogonal Transform Matrix

- Although a fixed randomized Hadamard matrix can reduce spike saturation, **layer-wise input distributions vary significantly**, making a fixed transform suboptimal.
- We therefore **initialize the transform with a randomized Hadamard matrix** and make it learnable, allowing each layer to adapt the transform to its own feature distribution.
- To preserve orthogonality, which enables RMSNorm equivariance and inference-time weight fusion, **we project the learnable matrix onto the nearest orthogonal matrix using the matrix sign function**. In practice, we approximate this projection with an efficient and differentiable fifth-order **Newton-Schulz iteration**, avoiding expensive SVD while enabling end-to-end optimization.



Experimental Results

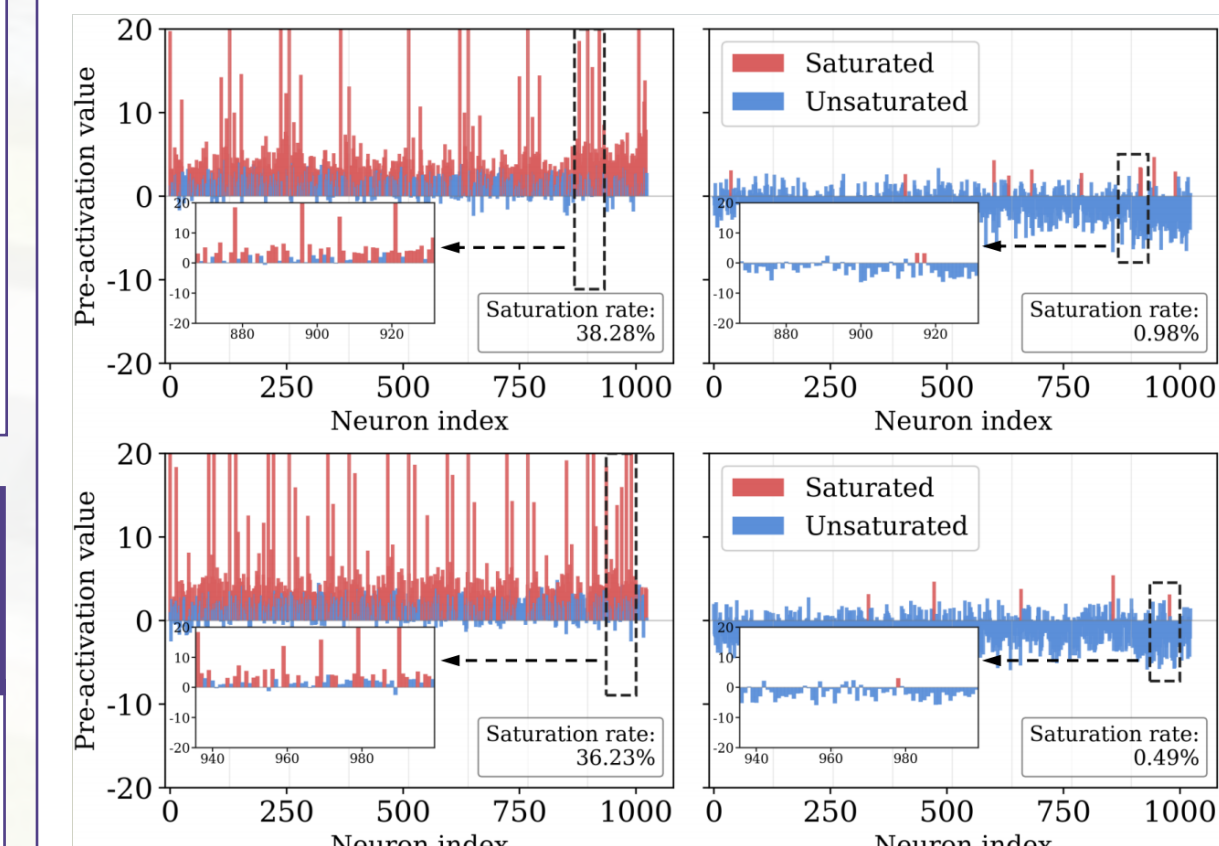
GLUE benchmark

Model	T	Energy	MNLI-m/mm	QQP	QNLI	SST-2	CoLA	STS-B	MRPC	RTE	Avg.
BERT _{base}	–	51.41	83.8/83.4	90.5	90.7	92.3	60.0	89.4	89.8	69.3	83.2
ELMo	–	–	68.6/–	86.2	71.1	91.5	44.1	70.4	76.6	53.4	70.2
BiBERT	–	–	66.1/67.5	84.8	72.6	88.7	25.4	33.6	72.5	57.4	63.2
BiT	–	–	77.1/77.5	82.9	85.7	87.7	25.1	71.1	79.7	58.8	71.0
BiPFT	–	–	69.5/70.6	83.7	81.7	86.2	22.9	80.2	76.2	66.1	70.8
SpikeBERT	4	14.30	71.4/71.0	68.2	66.4	85.4	16.9	18.7	82.0	57.5	59.7
PSN-BERT	4	–	35.4/35.2	0.0	50.5	50.9	0.0	6.8	81.2	52.7	34.7
LIF-BERT	4	7.98	56.8/55.2	70.0	60.6	80.6	14.6	20.0	82.3	53.8	54.9
Spikingformer	4	6.76	71.9/72.5	84.7	76.0	87.2	24.4	54.5	79.7	55.6	66.8
+SmoothSpike	4	9.45	75.1/76.1	87.6	83.8	88.6	41.2	79.3	84.8	58.5	75.0
SpikeLM	1	3.98	76.0/76.9	84.0	84.9	86.5	37.9	84.3	85.6	65.3	75.7
+SmoothSpike	1	7.03	76.8/77.7	84.3	86.8	89.5	52.7	83.5	88.1	58.1	77.5

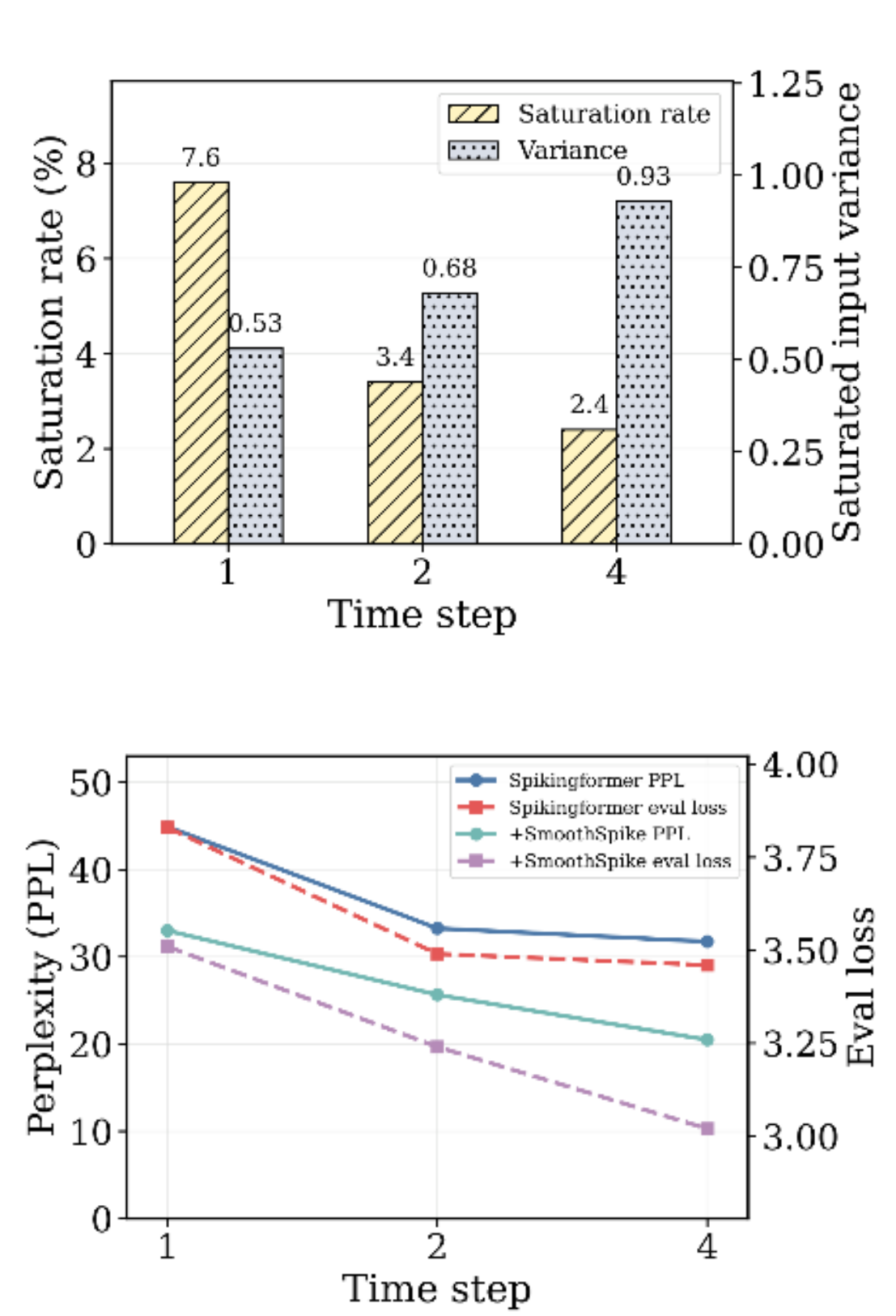
Ablation study

Method	H ₁	H ₂	H ₃	Learnable	MNLI-m/mm	QQP	QNLI	SST-2	CoLA	STS-B	MRPC	RTE	Avg.
✓	✓	✓	✓	✓	67.13/67.37	76.82	70.73	83.72	19.10	51.84	82.27	57.04	64.00
✓	✓	✓	✓	✓	67.83/68.45	78.19	71.13	83.03	19.24	50.55	81.48	53.79	63.74
✓	✓	✓	✓	✓	70.21/69.69	79.45	75.51	84.75	16.07	61.29	81.79	54.15	65.88
✓	✓	✓	✓	✓	71.16/71.30	80.24	77.06	86.01	22.30	64.18	82.04	54.51	67.64
✓	✓	✓	✓	✓	71.20/71.40	80.92	78.58	85.55	19.89	65.26	82.15	55.96	67.88

Effect of SmoothSpike



Effect of Time Steps



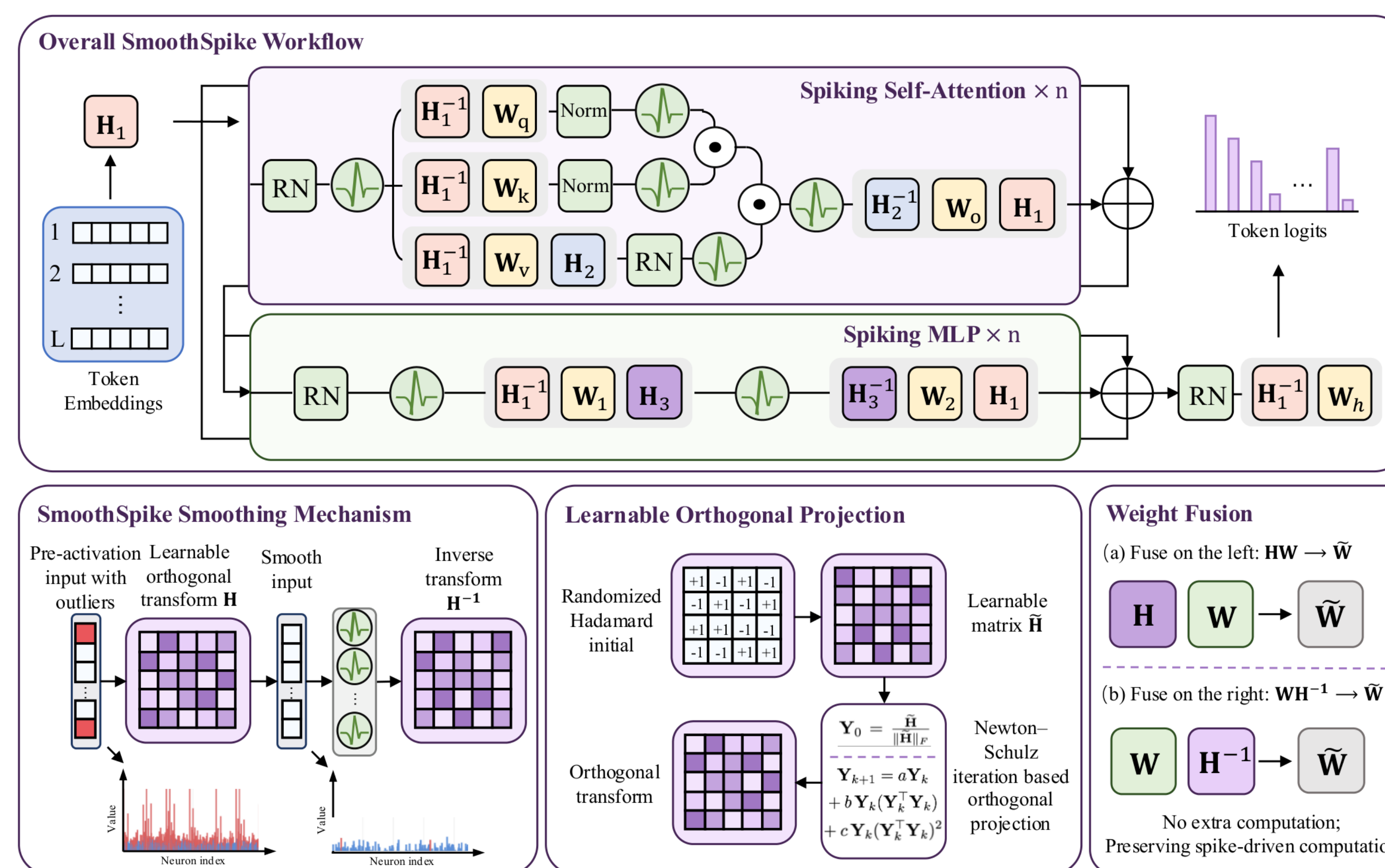
Conclusion

- We identify **spike saturation-induced information homogenization** as a key bottleneck that limits the representational capacity of SNNs.
- To address this, we use **learnable orthogonal transformations** to smooth extreme activations, with theoretical guarantees for suppressing maximum input magnitudes.
- Experiments show that SmoothSpike improves SNN performance while preserving spike-based energy efficiency.

Mitigating Spike Saturation

- To mitigate spike saturation, we apply a **randomized Hadamard transform** to the pre-activation inputs of spiking neurons, defined as $\tilde{x}^\ell = H_R x^\ell$, where H_R is an orthogonal Hadamard matrix with random sign flipping. This transform preserves the ℓ_2 norm of the input while redistributing its energy across all dimensions, preventing large values from being concentrated in a few neurons. Specifically, each transformed component is formed by $1/\sqrt{n}$ -scaled signed combinations of all input dimensions, where the positive and negative coefficients enable partial cancellation of extreme activations.
- After spike generation, we apply the inverse transform $y^\ell = H_R^T s^\ell$ to map the spike outputs back to the original representation space, forming an encoding-decoding pair that can be seamlessly inserted into existing SNN layers without disrupting information flow. Theoretically, the maximum transformed activation is bounded by $O(\sqrt{\log n / n})$ with high probability, showing that the transform effectively suppresses outlier inputs before they trigger saturation. Therefore, randomized Hadamard transformation reduces spike saturation, alleviates information homogenization, and helps preserve fine-grained semantic representations in deep spiking networks.

Overall Architecture



SmoothSpike augments spiking transformer blocks with learnable orthogonal transforms that smooth neuron inputs before spike generation and restore representations afterward. RMSNorm ensures orthogonal equivariance, allowing these transforms to be absorbed into linear weights during inference without changing the original event-driven computation.